

04 — Ordinal Modeling, Evaluation & Uncertainty

Measure whether the model is actually useful, not merely accurate

Build guide for the Context-Aware Online Distress Severity Detection project.

1. The Severity Spectrum Is Ordered

The classes have a natural order: Control → Low Stress → Moderate Distress → Severe Crisis.

A mistake between adjacent categories is conceptually different from confusing Control with Severe Crisis.

After establishing the ordinary multiclass baseline, investigate ordinal classification as an advanced experiment.

2. Evaluation Metrics

Primary metric: Macro F1, because it gives each class equal importance despite imbalance.

Safety-oriented metric: recall/sensitivity for the Severe Crisis class.

Also report precision, recall, F1 for every class, weighted F1, and confusion matrices.

AUROC and AUPRC can be considered where their multiclass interpretation is appropriate.

Never report only accuracy.

3. Example Results Table

```
Precision Recall F1
Control 0.XX 0.XX 0.XX
Low Stress 0.XX 0.XX 0.XX
Moderate Distress 0.XX 0.XX 0.XX
Severe Crisis 0.XX 0.XX 0.XX
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Macro 0.XX 0.XX 0.XX
```

4. Important Rule About the Claimed 0.89 / 94% Results

The original description states 0.89 Macro F1 and 94% acute sensitivity. Those should be treated as target or historical claims until reproduced under a documented dataset split and evaluation protocol.

Do not put these numbers on a resume or final report unless your actual experiments produce them and you can explain the exact evaluation setup.

5. Uncertainty and Calibration

A softmax probability is not automatically a trustworthy confidence estimate.

For an example prediction, the system can expose a probability distribution across the four classes.

Then test whether those probabilities are calibrated.

Temperature scaling is a practical post-training calibration method: raw logits are adjusted by a learned temperature before softmax probabilities are interpreted.

6. Calibration Evaluation

Use Expected Calibration Error (ECE) and reliability diagrams.

Compare the uncalibrated and calibrated model.

The goal is not simply to make probabilities look less confident; it is to make confidence correspond more closely to empirical correctness.

7. Attention Analysis

Extract self-attention information from selected transformer layers and heads.

Visualize attention patterns with PyTorch and Matplotlib.

Use this as attention-based model introspection.

Important caveat: attention weights should not automatically be described as faithful explanations or proof that specific words caused a prediction.

8. Stronger Interpretability

For a more defensible explanation layer, investigate Integrated Gradients, SHAP where appropriate, and token occlusion/perturbation.

Compare attribution results with attention visualizations instead of assuming they are equivalent.

9. Error Analysis

Create explicit buckets for false positives and false negatives.

Useful categories include sarcasm, slang, figurative language, negation, implicit distress, context dependence, ambiguous language, and annotation disagreement.

False negatives in the Severe Crisis class deserve especially careful inspection because they represent the failure mode the loss design is trying to reduce.